Exp no 1: Simple Reflex Agent for Vacuum Cleaner World

Experiment Title

Design and Implementation of a Simple Reflex Agent for the Vacuum Cleaner World.

Aim

To design and implement a Simple Reflex Agent for the Vacuum Cleaner World and analyze its rational behavior in a given environment.

Theory

A Simple Reflex Agent selects actions based only on the current percept. In the Vacuum Cleaner World, the agent perceives its current location and whether the room is clean or dirty. If the room is dirty, the agent cleans it; otherwise, it moves to the other room. The agent is considered rational because it chooses actions that help keep the environment clean. Artificial Intelligence (AI) is a branch of computer science that focuses on creating systems capable of performing tasks that normally require human intelligence, such as reasoning, learning, problem-solving, perception, and decision-making. An intelligent agent is an entity that perceives its environment through sensors and acts upon that environment through actuators to achieve specific goals.

A **Simple Reflex Agent** is the most basic type of intelligent agent. It selects actions based solely on the current percept and ignores the history of previous percepts. Such agents use a set of predefined **condition-action rules**, also known as production rules. When a particular condition is satisfied, the corresponding action is executed immediately.

The **Vacuum Cleaner World** is one of the simplest environments used in Artificial Intelligence to demonstrate the behavior of intelligent agents. The environment consists of two rooms, Room A and Room B. Each room can be either **Clean** or **Dirty**. The vacuum cleaner agent can perceive:

- Its current location (Room A or Room B).
- The status of the room (Clean or Dirty).

Based on these percepts, the agent can perform one of the following actions:

1. **Suck** – Clean the current room if it is dirty.
2. **Move Left** – Move to the left room.
3. **Move Right** – Move to the right room.

The Simple Reflex Agent follows the rules:

- If the current room is dirty, perform the **Suck** action.
- If the room is clean, move to the other room.

The primary goal of the agent is to keep both rooms clean while minimizing unnecessary actions. The agent is considered **rational** because it always chooses the action that maximizes its performance measure, which is maintaining a clean environment.

The Vacuum Cleaner World helps students understand fundamental AI concepts such as:

- Intelligent Agents
- Rational Agents
- Agent Functions
- Percepts and Actions
- Environment Representation
- Problem Solving in AI

Although Simple Reflex Agents are easy to design and implement, they have certain limitations. They do not maintain an internal state and cannot remember previous actions. Therefore, they work effectively only in fully observable and simple environments. For more complex environments, advanced agents such as **Model-Based Agents**, **Goal-Based Agents**, and **Utility-Based Agents** are preferred.

Applications

1. Robotic vacuum cleaners.
2. Automated cleaning systems.
3. Intelligent monitoring systems.
4. Basic AI decision-making systems.
5. Autonomous agents in simple environments.

Limitations

- Does not remember past states.
- Cannot handle partially observable environments.
- Not suitable for complex problems.
- May perform repetitive actions.

Viva Questions(write any 5 Questions)

1. What is Artificial Intelligence?
2. Define an Intelligent Agent.
3. What is a Simple Reflex Agent?
4. What are percepts and actions?
5. Why is the Vacuum Cleaner World used in AI?
6. What makes an agent rational?
7. What is the PEAS representation?
8. What are the limitations of a Simple Reflex Agent?
9. Differentiate between Simple Reflex Agent and Model-Based Agent.
10. What is an environment in AI?
11. Explain the Suck action in the Vacuum Cleaner World.
12. What are the characteristics of a rational agent?
13. What is an agent function?
14. What is an agent program?
15. Name different types of intelligent agents.

Python Code

```
# Simple Reflex Agent for Vacuum Cleaner World

def vacuum_agent(location, status):
    if status == "Dirty":
        return "Suck"
    elif location == "A":
        return "Move Right"
    else:
        return "Move Left"

rooms = {"A": "Dirty", "B": "Clean"}
location = "A"

for i in range(4):
    print("Location:", location, "| Status:", rooms[location])

    action = vacuum_agent(location, rooms[location])
    print("Action:", action)

    if action == "Suck":
        rooms[location] = "Clean"
    elif action == "Move Right":
```

```
        location = "B"
    elif action == "Move Left":
        location = "A"

    print()
```

Output

```
Location: A | Status: Dirty
Action: Suck

Location: A | Status: Clean
Action: Move Right

Location: B | Status: Clean
Action: Move Left

Location: A | Status: Clean
Action: Move Right
```

Conclusion

The Simple Reflex Agent successfully keeps the environment clean by following condition-action rules. This demonstrates rational behavior in the Vacuum Cleaner World.